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Review

Applications of machine learning tools for biological treatment of organic wastes: Perspectives and challenges

Long Chen [a, b], Pinjing He [a, b, c], Hua Zhang [a, b, c], Wei Peng [a, b, c], Junjie Qiu [a, b, c], Fan Lü [a, b, c, *]

[a] State Key Laboratory of Pollution Control and Resource Reuse, College of Environmental Science and Engineering, Tongji University, Shanghai 200092, China
[b] Institute of Waste Treatment and Reclamation, Tongji University, Shanghai 200092, China
[c] Shanghai Institute of Pollution Control and Ecological Security, Shanghai 200092, China

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ABSTRACT

Biological treatment technologies (such as anaerobic digestion, composting, and insect farming) have been extensively employed to handle various degradable organic wastes. However, the inherent complexity and instability of biological treatment processes adversely affect the production of renewable energy and nutrient-rich products. To ensure stable processes and consistent product quality, researchers have invested heavily in control strategies for biological treatment, with machine learning (ML) recently proving effective in optimizing treatment, predicting parameters, detecting disturbances, and enabling real-time monitoring. This review critically assesses the application of ML in biological treatment, providing an in-depth evaluation of key algorithms. This study reveals that artificial neural networks, tree-based models, support vector machines, and genetic algorithms are the leading algorithms in biological treatment. A thorough investigation of the applications of ML in anaerobic digestion, composting, and insect farming underscores its remarkable capacity to predict products, optimize processes, perform real-time monitoring, and mitigate pollution emissions. Furthermore, this review outlines the challenges and prospects encountered in applying ML to biological treatment, highlighting crucial directions for future research in this area.

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1. Introduction

Biological technologies, such as anaerobic digestion (AD), composting, and insect farming, involve the use of organisms or microorganisms to treat organic waste and recover energy, organic fertilizers, proteins, lipids, and other valuable products (Puyol et al., 2017; Sharma et al., 2020). They have garnered widespread attention because they effectively render organic waste harmless, stable, and reduced in volume, converting it into resources with lower investment and operational costs compared with physical and chemical methods (Hoang et al., 2022; Lohri et al., 2017). However, biological processes are inclined to exhibit complexities and instabilities because of various factors such as microbial growth rates, substrate concentration, temperature, and pH, adversely affecting the efficiency of renewable energy and nutrient-rich product production (Leitão et al., 2006; Sipma et al., 2010). Moreover, improper management of byproducts from abnormal operations can lead to secondary pollution, adversely affecting the environment (Guo et al., 2022; Yuan & Zhu, 2016). Thus, it is crucial to implement effective measures for prediction, monitoring, and control to ensure a more economical, stable, and efficient biological process. Although some conventional mathematical models offer various statistical evaluations of variable values, their predictive performance is constrained by assumptions, data requirements, uncertainties, and single-factor analysis (Rigdon, 2012; Tedeschi, 2006). Variable transformations can be employed to improve the performance of conventional mathematical models, but suboptimal results may still be obtained (Aydın Temel et al., 2023). Therefore, when predicting and optimizing complex biological treatment processes, it is necessary to adopt other more advanced methods and tools to enhance the accuracy and reliability of prediction and optimization.

Machine learning (ML), as a crucial branch of artificial intelligence, has garnered widespread acclaim because of its time

* Corresponding author. State Key Laboratory of Pollution Control and Resource Reuse, College of Environmental Science and Engineering, Tongji University, Shanghai 200092, China.
E-mail address: lvfan.rhodea@tongji.edu.cn (F. Lü).

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savings, high prediction accuracy when facing intricate nonlinear issues, and remarkable reductions in labor and resource consumption through the elimination of redundant experiments (Guo et al., 2021a; Khan et al., 2023). These characteristics enable ML to overcome the limitations of conventional biological treatment methods. Dominant ML methodologies include supervised learning (such as classification and regression), unsupervised learning (such as clustering), and reinforcement learning (Bertolini et al., 2021). In the field of biological treatment, supervised ML algorithms, including artificial neural networks (ANNs) (Zhan & Zhu, 2024), tree-based models (Li et al., 2022a), support vector machine (SVM) (Ge et al., 2023), the genetic algorithm (GA) (Awhangbo et al., 2024), and various multivariate statistical analysis (MSA) methods (Gil et al., 2018), have been used to model nonlinear and complex relationships.

First, ML has been employed to predict the yield and quality of biological treatment. For instance, it has accurately forecasted AD biogas production and composition across laboratory and industrial scales (Li et al., 2022a; Sappl et al., 2023), anticipated compost maturity (Kujawa et al., 2020), and predicted expected weights in black soldier fly (BSF) farming (Muinde et al., 2023). Moreover, by integrating optimization algorithms such as the GA and particle swarm optimization (PSO), ANNs have been used to optimize process parameters in various biological treatment technologies, such as AD (Dahunsi et al., 2016), composting (Ding et al., 2022), and BSF farming (Proietti et al., 2022). ML has also been used to monitor and control the stability of biological treatment and pollution emissions. For example, algorithms such as ANNs and MSA have been used to monitor and regulate key indicators in the AD process (Jia et al., 2022), oversee pollution and greenhouse gas emissions in composting processes (Guo et al., 2022), and manage the lifecycle of BSFs using computer vision-based algorithms (Proietti et al., 2022). However, despite its vast potential in biological treatment, ML still faces challenges, including complex parameter settings, poor model generalization, high computational resource consumption, and lack of explainability. Addressing these challenges will significantly propel the further application of ML in the field of biological treatment.

Until now, several review articles have discussed the application of ML, particularly in AD (Cruz et al., 2022; Khan et al., 2023) or composting (Aydın Temel et al., 2023). However, a comprehensive review focusing on the application of ML in biological treatment is still lacking. The prevalent patterns, common challenges, and prospects of ML applications within the biological treatment domain necessitate further exploration and elucidation. Therefore, this study provides a systematic and up-to-date review of the application of ML in biological treatment. The framework of this review is structured as follows: In Section 2, a brief introduction is provided to various popular ML algorithms used in this field, along with a comparison of their advantages, disadvantages, and applicable scopes. Sections 3, 4, and 5 discuss studies on ML for modeling the processes of AD, composting, and insect-based waste treatment, respectively. Before summarizing the findings of this study in Section 7, the current challenges in successfully deploying ML in biological treatment and future perspectives are outlined in Section 6.

2. Development trends and typical algorithms for ML applications in biological treatment

The fundamental principles and workflow of ML are illustrated in Fig. S1 which is in the Electronic Supplementary Material (ESM) in the online version of this paper. Based on data retrieved from the Web of Science (Fig. 1(a)), the number of studies investigating the use of ML in the biological treatment of organic waste has steadily increased from 2017 to 2023. However, the overall volume of literature remains limited, indicating significant development potential for the applications of ML in biological treatment. Regarding the proportion of publications on different treatment approaches (Fig. 1(b)), the order of percentage is AD (65.50%) > composting (32.17%) > insect farming technology (2.33%). Figs. 1(c) and 1(d) show the most commonly employed ML algorithms in modeling biological treatment, which are categorized into five primary types: ANNs, tree-based models (RF/DT/GBR), SVM, GA, and others. Among them, ANN models are the most commonly used algorithms. Specifically, research employing ANNs accounts for 32.65% and 33.33% of the total AD and composting studies, respectively. Following ANNs, tree-based models feature significantly, accounting for 22.96% and 26.26% in AD and composting studies, respectively. The use rates of SVM and GA models are lower, at 14.80% and 18.18% for AD and 12.76% and 10.10% for composting, respectively. Apart from these models, other models demonstrate application rates of 16.83% and 12.13%, respectively. Table S1 in the ESM provides a detailed summary of the advantages, disadvantages, and suitable application domains of these algorithms. Notably, there remains a scarcity of research addressing ML applications in insect farming technology, which can be attributed to limited engineering practices to date. This topic will be discussed in Section 5 (ML applications for insect treatment of organic waste).

2.1. Artificial neural networks (ANNs)

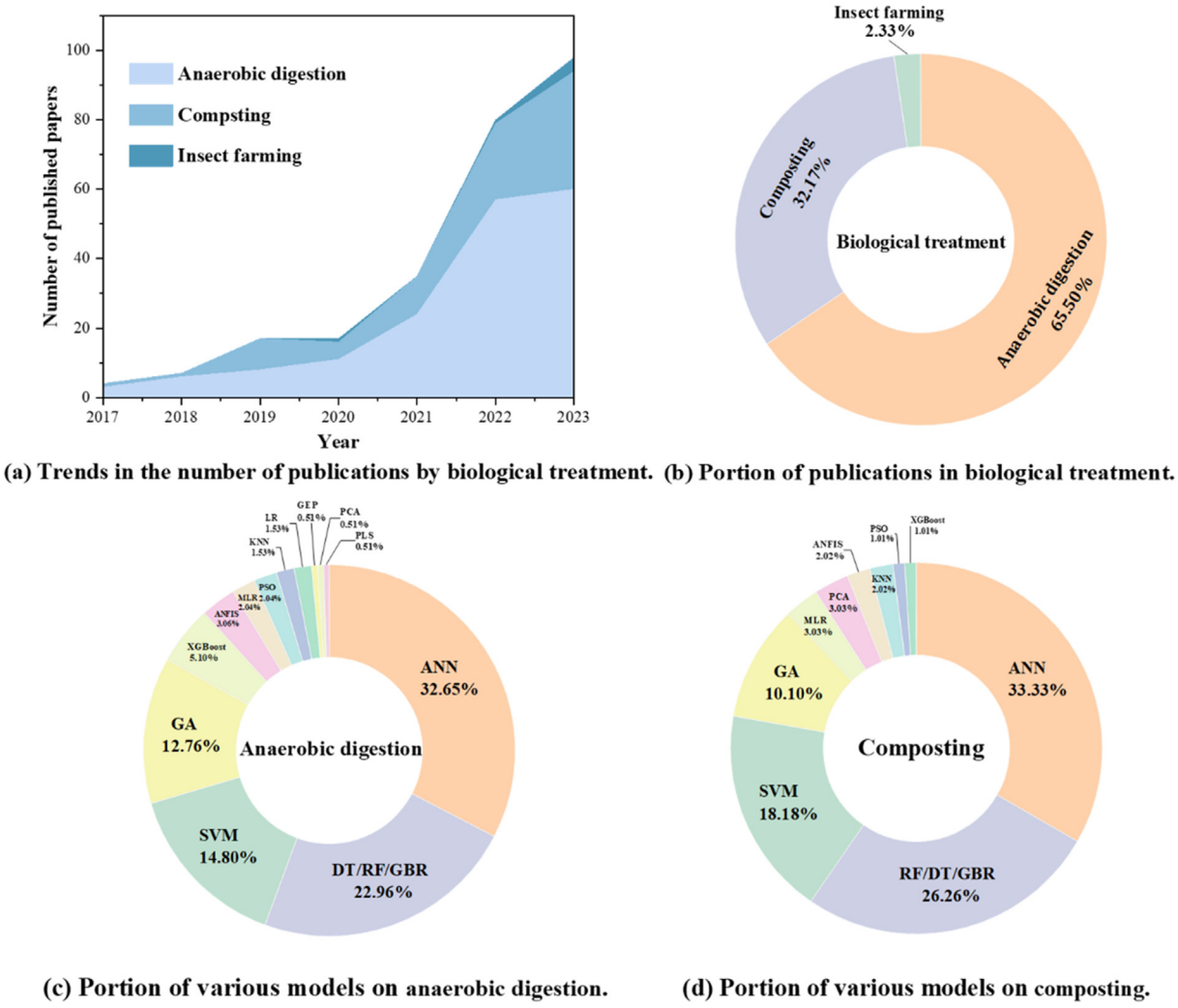
ANNs emulate the functionality of neurons in the human brain and represent a critical technology in the field of ML (Aghbashlo et al., 2021). They exhibit robust modeling capabilities for complex nonlinear systems by discerning inherent patterns and hierarchical representations within extensive input data. In recent years, particularly with the advent of deep neural networks (DNNs), the performance of ANNs has significantly improved, leading to breakthroughs in image recognition, data processing, and other areas. In biological treatment research, a range of ANN models, such as multilayer perceptron (MLP), recurrent neural networks (RNNs), long short-term memory (LSTM), and convolutional neural networks (CNNs), have recently been extensively employed.

2.2. Tree-based models

Tree-based models use tree structures to iteratively divide datasets into distinct subsets, employing tailored decision rules for each subset. This approach ultimately leads to accurate decision making and prediction. Among various tree-based models, decision tree (DT), random forest (RF), and gradient boosting regression (GBR) are the most widely used in biological treatment owing to their remarkable flexibility and adaptability in handling intricate biological data.

2.3. Support vector machine (SVM)

The core idea of SVM is to find an optimal hyperplane in a high-dimensional space to maximize the distance from it to the nearest data points, known as support vectors or maximum-margin hyperplane (Pisner & Schnyer, 2020). SVM employs kernel methods to achieve optimal data separation and follows the structural risk minimization principle (Gupta & Gupta, 2023). SVM has exhibited remarkable predictive performance in organic waste classification, AD, and compost quality assessment (Aydın Temel et al., 2023; Ge et al., 2023). Furthermore, when SVM is used for regression analysis in research, it is referred to as support vector regression (SVR).



(a) Trends in the number of publications by biological treatment. (b) Portion of publications in biological treatment.

(c) Portion of various models on anaerobic digestion. (d) Portion of various models on composting.

Fig. 1. Trends and proportion of publications on biological treatment applications (ANNs: artificial neural networks; DT: decision tree; RF: random forest; GBR: gradient boosting regression; SVM: support vector machine; GA: genetic algorithm; XGBoost: extreme gradient boosting; MLR: multiple linear regression; PSO: particle swarm optimization; KNN: K-nearest neighbor; ANFIS: adaptive network-based fuzzy inference system; LR: linear regression; GEP: gene expression programming; PLS: partial least square; PCA: principal component analysis).

2.4. Nature-inspired optimization algorithms

The most commonly used nature-inspired optimization algorithms in biological treatment are GAs and PSO. GAs mimic natural selection and genetic processes in biological evolution, evolving superior solutions via selective breeding, genetic crossover, and mutation within a randomized solution pool (Renner & Ekárt, 2003). Conversely, PSO emulates animal swarming behavior, where each solution is a particle that updates its position and velocity by following its personal best and the global best, converging toward the ultimate optimal solution (Pornsing et al., 2016).

2.5. Other algorithms

2.5.1. Multivariate statistical analysis (MSA)

Principal component analysis (PCA), multiple linear regression (MLR), and partial least squares (PLS) are the most prevalent MSA tools for modeling biological treatment. PCA uses orthogonal transformations to convert a set of potentially correlated variables into linearly independent components, thereby aiding in pinpointing the key factors affecting the outcome variable (Jolliffe & Cadima, 2016). MLR examines relationships between multiple variables; however, for nonlinear associations, alternative nonlinear regression methods are crucial for improved accuracy (Krmar et al., 2020). PLS regression, as a highly optimized regression technique, integrates the core principles of PCA and MLR, demonstrating superior performance in handling intricate data (Chen et al., 2022a). Unlike PCA, PLS further delves into feature interrelationships, thereby improving the accuracy of capturing complex data patterns and latent connections (Kettaneh et al., 2005).

2.5.2. Boosting algorithms

Adaptive boosting (AdaBoost) and extreme gradient boosting (XGBoost) are among the most commonly used boosting algorithms in the biological treatment of organic waste. AdaBoost assigns weights to each training sample, giving greater importance to misclassified samples during iterations, making it effective for both classification and regression tasks (Ferreira & Figueiredo, 2012). Conversely, XGBoost employs gradient boosting trees, incorporating regularization terms and a learning rate during the construction of each tree to control model complexity and improve

generalizability (Shehadeh et al., 2021). Furthermore, XGBoost supports parallel computing and allows the customization of optimization objective functions, exhibiting excellent performance on large-scale datasets.

2.5.3. K-nearest neighbor (KNN)

K-nearest neighbor (KNN) is an instance-based learning algorithm that classifies or predicts new data points by comparing them to the K-nearest neighbors in the training dataset using their categories or attribute values (Uddin et al., 2022). KNN does not require explicit model training but directly uses existing instances for predictions, which makes it efficient in handling large-scale datasets. However, KNN is sensitive to the local data structure, susceptible to noise, and has high computational complexity when dealing with high-dimensional data (Jiang et al., 2020). Furthermore, the selection of the K value plays a crucial role in determining the performance of KNN (Zhang et al., 2018).

2.5.4. Adaptive network-based fuzzy inference system (ANFIS)

The ANFIS integrates principles from neural networks and fuzzy logic, enabling high-precision simulation of nonlinear systems, such as predicting and optimizing the AD process in biogas production (Najafi & Ardabili, 2018; Zareei & Khodaei, 2017). This integrated approach combines the learning capabilities of neural networks with the enhanced decision making abilities of fuzzy inference systems to optimize model performance. In addition, ANFIS offers excellent explainability and transparency (Chen et al., 2022b). These characteristics make ANFIS highly promising and versatile for various applications in biological treatment.

3. ML applications in AD

AD is a sustainable bioprocessing method that uses specific microorganisms, such as methane-producing bacteria, to convert organic matter into high-energy biogas, primarily composed of CH_4 and CO_2 (Wu et al., 2021; Zamri et al., 2021). To overcome the challenges faced in AD, including sensitivity to feedstock composition and operating parameters and difficulties in predicting and controlling the stability and biogas production efficiency of AD systems, an increasing number of researchers are exploring the application of ML techniques for monitoring, controlling, and optimizing the AD process. ML algorithms, such as ANNs, RF, SVM, and GA, have been employed to predict biogas yield and composition, monitor and control process stability, and optimize process parameters, thereby enhancing the efficiency and quality of AD. The principles of AD research based on ML are presented in Fig. 2.

3.1. Biogas yield and composition prediction

3.1.1. Progress in laboratory-scale AD

ML models have found widespread application in predicting biogas production (Khashaba et al., 2022; Long et al., 2021; Mo et al., 2023; Neto et al., 2021; Okwu et al., 2021) and compositions (CH_4, CO_2, H_2) (Chong et al., 2023; Li et al., 2022b; Sun et al., 2024) in laboratory-scale AD. For instance, independent ANN models have been extensively employed to model AD processes, predicting biogas production and composition (Cruz et al., 2023; Ghatak & Ghatak, 2018; Jaroenpoj et al., 2014; Long et al., 2021; Neto et al., 2021). To further improve the predictive performance of ANN models for biogas production, combinations of ANN models with nature-inspired optimization algorithms have been used to select the optimal ANN structure (Beltramo et al., 2019; Gueguim Kana et al., 2012; Oloko-Oba et al., 2018; Zaied et al., 2023). For example, Zaied et al. (2023) and Machineni and Anupoju (2023) integrated ANNs with PSO to optimize ANN parameters (weights and biases), whereas other studies have employed the GA (Ghersi et al., 2021; Jacob & Banerjee, 2016; Zhan & Zhu, 2024), ant colony optimization (Fajobi et al., 2023), and the seagull optimization algorithm (Abdel daiem et al., 2022) for selecting the most effective ANN parameters in predicting biogas production. Furthermore, some studies have applied ANFIS to predict biogas production in AD, achieving R^2 values exceeding 0.95 (Fajobi et al., 2023; Heydari

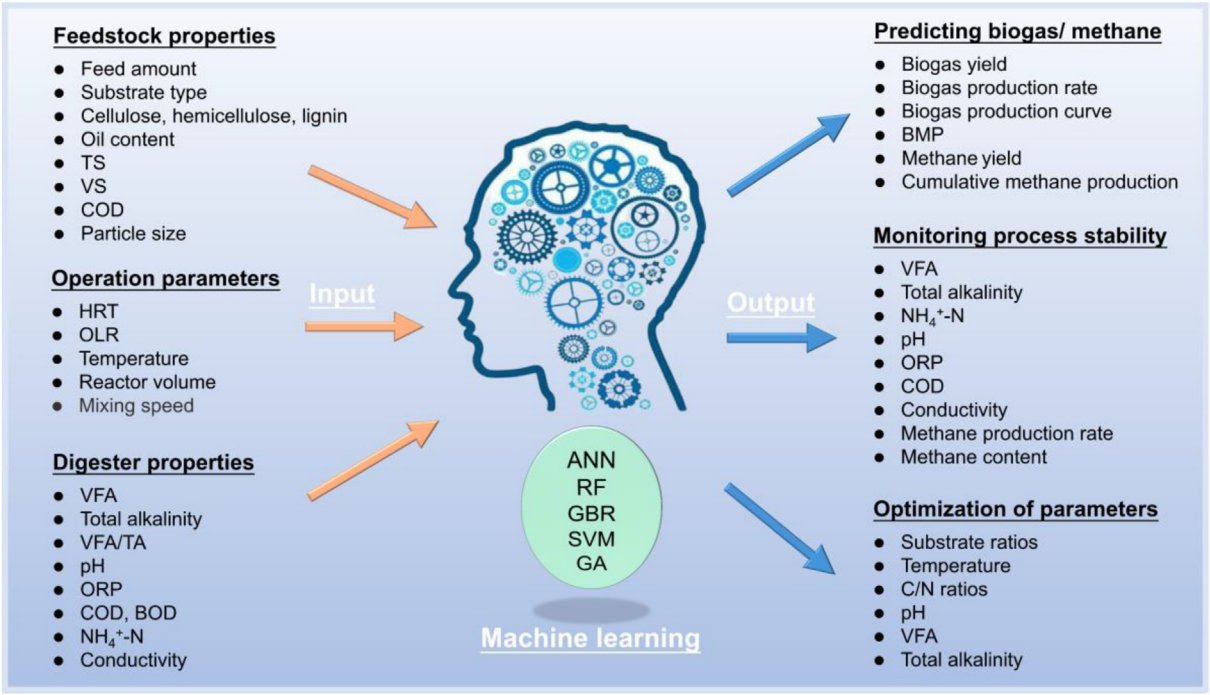


Fig. 2. Application principle of AD research based on ML.

et al., 2021; Okwu et al., 2021; Olatunji et al., 2022). Although ANN models have demonstrated superior performance in modeling AD systems, their main limitation lies in their "black-box" nature, which results in poorer explainability. Although neural networks often exhibit high predictive accuracy, they rely on extensive training data and hyperparameter optimization. In addition, challenges such as overfitting, substantial computational resource consumption, and insufficient generalization ability need to be addressed when using ANN models to predict biogas production in AD.

Compared with other ML methods, tree-based models are typically easier to implement and require less data preprocessing. Furthermore, these models allow users to easily extract the importance of features, thereby improving model performance, aiding in the construction of interpretable models, and enhancing our understanding of the current context by capturing relevant information within the model. Several studies have used tree-based models to predict biogas production and composition during AD processes (Chiu et al., 2022; Sun et al., 2024; Wang et al., 2021, 2023b). For instance, Zhang et al. (2023) developed RF, GBR, and XGBoost models to predict the facilitating effects of biochar on methane yield and the maximum methane production rate during AD. The GBR model achieved R^2 values of 0.84 and 0.69 for the methane yield and methane production rate, respectively. The optimal methane production and methane production rate occurred with biochar particles of 0.3−0.5 mm, a surface area of ~290 m^2/g, >31% oxygen, and >20 g/L biochar addition.

Other studies have also compared various ML algorithms for modeling the AD process (Chong et al., 2023; Heydari et al., 2021; Obileke et al., 2023; Okwu et al., 2023; Tufaner & Demirci, 2020). For example, Okwu et al. (2023) compared the Mamdani fuzzy model (MFM), ANNs, and response surface methodology (RSM) for modeling and optimizing methane production in AD and discovered that the predictive performance of the MFM and ANN models was slightly higher than that of the RSM model. Similarly, Heydari et al. (2021) evaluated the predictive performance of ANNs and ANFIS for methane production in an upflow anaerobic sludge blanket reactor, revealing that the ANN model exhibited slightly better predictive performance than ANFIS.

In summary, ANN models have become the preferred ML algorithm for predicting biogas production and composition in AD processes. However, despite the excellent predictive accuracy of ANN models, issues related to explainability, computational costs, and adaptability of neural network models have prompted the exploration of tree-based models. In particular, when handling complex systems such as AD, where understanding the relationships between variables is crucial, tree-based models have demonstrated their unique advantages. An overview of recent studies on the application of ML models in laboratory-scale AD is summarized in Table 1.

3.1.2. Progress in industrial-scale AD

In recent years, ML models applied to predict biogas production in industrial-scale AD has also made some strides (as summarized in Table 2). Specifically, Asadi et al. (2020) used ANFIS with fuzzy c-means clustering to predict biogas production rates (CH_4, CO_2, and H_2S) in AD tanks at a wastewater treatment plant. Wang et al. (2021) analyzed the impact of different types of waste and operational parameters on industrial-scale AD biogas production using various ML methods. The results indicated that although the changes in biogas production were mainly attributed to diverse waste types, the predictive capability of the operational parameters was comparatively limited. Li et al. (2022a) employed XGBoost, SVM, ANN, RF, and linear regression (LR) to forecast biogas production within an industrial-scale biogas facility. Their findings showed that the RF model performed best, with an average R^2 of 0.74. Feature importance analysis revealed that feedstock amount, digestate properties, and feedstock properties were significant factors, accounting for 45.9%, 38.6%, and 15.4%, respectively. Similarly, Yildirim and Ozkaya (2023) used RF, ANN, KNN, SVR, and XGBoost models to forecast biogas production in a large-scale AD facility. Among these methods, the RF model emerged as the most accurate predictor, attaining a high R^2 of 0.9242. Overall, existing studies have underscored the potential of ML algorithms for forecasting biogas production on an industrial scale. However, most current applications mainly focus on utilizing existing operational data to forecast biogas production or composition, leaving several avenues for further exploration and enhancement, such as developing real-time monitoring and control systems and integrating other advanced technologies such as the Internet of Things and cloud computing.

3.2. Monitoring and control

The accumulation of inhibitory compounds, resulting from various factors, including high substrate concentrations, pH fluctuations, unsuitable temperatures, and excessive organic load, frequently causes reactor failure in the AD process (Wu et al., 2019; Xie et al., 2016). Therefore, significant attention has been paid to the development of process monitoring and control technologies to maintain process stability and maximize methane production. However, conventional AD process monitoring methods typically involve extensive and time-consuming chemical and physical analysis, which are not only costly and imprecise but also challenging to implement for real-time monitoring (Waltham & Örmeci, 2020; Wang et al., 2018). Consequently, automated and intelligent control strategies, particularly those using prediction control and optimization algorithms based on ML, have emerged as viable solutions for real-time monitoring, optimization, and control of the AD process by leveraging input variables and their hidden interdependencies (Khan et al., 2023; Nguyen et al., 2015). Various offline and online indicators have been commonly used to monitor AD process stability, including biogas yield and composition, pH, chemical oxygen demand (COD), oxidation−reduction potential (ORP), ammonia/ammonium (NH_3/NH_4^+), volatile fatty acids (VFA), alkalinity (ALK), VFA/ALK, concentrations of specific intermediate metabolites (CO, H_2, aromatic compounds, etc.), microbial community, and microbial activity (Cruz et al., 2021; Khan et al., 2023; Nie et al., 2023). Among these indicators, VFA, ALK, COD, and pH are the most commonly monitored. Furthermore, ANNs and MSA are the most widely used ML models for monitoring these indicators.

VFA is the intermediate product resulting from the degradation of organic matter in AD. The accumulation of excessive VFA concentrations can inhibit microbial populations within the digester, making it a crucial indicator requiring close monitoring in the AD process (Harirchi et al., 2022). To overcome the limitations posed by expensive and environmentally sensitive physical sensors for VFA measurements, Kazemi et al. (2020) proposed the use of data-driven soft sensors that rely on ML techniques for reliable monitoring and prediction of VFA levels. Compared with ANNs, RF, SVM, and extreme learning machines, genetic programming soft sensors exhibit more precise predictive performance and better robustness. Furthermore, to further enhance the predictive capability of soft sensors, Xu et al. (2020) developed an integrated simulation method based on random standard deviation sampling and DNNs to predict VFA production in AD processes. The optimum R^2 was determined to be 0.998, with the smallest average absolute error of 3.28%.

ALK is another pivotal indicator that requires close monitoring within the AD process. Ward et al. (2011) were pioneers in developing a software sensor based on MLR to predict the ALK of AD tanks using pH, electrical conductivity (EC), and ORP as input

Table 1
The latest applications of ML in laboratory-scale AD to predict biogas production.

Substrates	Input variable	Output variable	ML algorithms	Best performance (R^2, RMSE, MSE, MAE)	Reference
Cattle dung, bamboo dust, sugar-cane bagasse, saw dust	Composition, temperature, time, and HRT	Biogas production curve	ANN	$R^2 = 0.9984$, AARE = 0.043%	Ghatak and Ghatak (2018)
Anaerobic hybrid reactor (AHR), perlite	Reactor fill ratio, influent pH, effluent pH, influent alkalinity (ALK), effluent ALK, OLR, effluent COD, effluent TSS, effluent SS and effluent VSS	Biogas production rate	ANN, Nonlinear regression models	ANN; $R^2 = 0.9852$, RMSE = 217.4	Tufaner and Demirci (2020)
Agricultural wastes, cow manure	Temperature, chemical dose, composition, and HRT	Cumulative methane production	ANN	$R^2 = 0.985$, RMSE = 0.0035	Almomani (2020)
Crop straw, livestock manure	Crude protein, crude fat, cellulose, hemicellulose, lignin, total sugar, C/N	Cumulative methane yield	MVLR, MCARS, GSA, PLS	MCARS; $R^2 = 0.982$, RMSE = 6.599	Yang et al. (2021a)
Food, fruits and vegetables wastes	Mixing ratios, reactor/feed type, VS, pH, OLR, HRT, Temperature, reactor volume	Cumulative biogas production	ANN	$R^2 = 0.8486$	Neto et al. (2021)
Spearmint essential oil wastewater	COD, pH, effluent MLSS, MLVSS, turbidity, oil and grease removal, COD removal, phenol removal, effluent volatile fatty acids (VFA) and ALK	Biogas production	ANN, ANFIS	ANN; $R^2 = 0.975$, RMSE = 2650, relative root mean = 0.234%	Heydari et al. (2021)
Poultry wastes and cow dung	Mixing ratio, HRT	Biogas yield	ANFIS, RSM	ANFIS; $R^2 = 1$, RMSE = 1.0, MAE = 0	Okwu et al. (2021)
Arachis hypogea shells	Temperature, HRT, and particle size	Biogas and methane yields	ANFIS, RSM, FCM	ANFIS; $R^2 = 0.9601$, 0.9486, 0.9626, 0.9172	Olatunji et al. (2022)
Manure, food waste, sewage sludge	Operating mode, HRT, Temperature, additives, co-substrate	Antibiotic resistance genes, mobile genetic elements	ANN, RF, XGBoost	XGBoost; $R^2 = 0.9$	Haffiez et al. (2022)
Lignocellulosic biomass	VS/TS, cellulose, hemicellulose, lignin, soluble saccharide, and crude protein. C, H, O, N	Biochemical methane potential (BMP)	ANN, DT, ELM, EN, GPR, KNN, MLR, PR, SVM, PCA	KNN; $R^2 = 0.75$, MAE = 30.2	Wang et al. (2023b)
Palm oil mill effluent	COD, BOD, TS, OLR, HRT, VFA, recirculation ratio, pH and temperature	Biogas production and methane yield	RSM, ANN, ANFIS	ANFIS; $R^2 = 0.9791$, MAE = 0.0730, RMSE = 0.1438	Chong et al. (2023)
Cow dung	Global solar irradiance, absolute relative humidity, slurry temperature, biogas temperature, absolute pH, and ambient temperature	Biogas yield	ANN, MLR	ANN; $R^2 = 0.999$, RMSE = 8.33×10^{-6}	Obileke et al. (2023)
Poultry wastes and cow dung	The ratio of cow dung to poultry wastes, and the retention time	Biogas yield	MFM, BP-ANN, RSM	MFM, ANN; $R^2 = 1.0$	Okwu et al. (2023)
Biochar, food waste, manure, and other biomass	Physicochemical properties of biochar, material conditions of AD, operating conditions for AD	Methane yield	GBR, RF, XGBoost	GBR; $R^2 = 0.84$, RMSE = 72.04	Zhang et al., (2023)
Cassava wastewater, sewage sludge	pH, calcium concentration and reaction time	Biogas production	ANN, GA	$R^2 = 0.9999$	Cruz et al. (2023)
Sweet sorghum leaves and stalks, mixed microbial consortia	Mixing speed, pH, reaction temperature, and SIRs	BMP	ANN, PSO	$R^2 = 0.996$, RMSE = 308.42	Machineni and Anupoju (2023)
Agricultural waste	Biomass type, reactor/feeding, VS, pH, OLR, HRT, temperature, and reactor volume	Steady-state biogas production	ANN, COA, MVO, LCA, ERWCA, SFS, TLBO	ANN-ERWCA; $R^2 = 0.9314$, RMSE = 0.1969, MAE = 0.1307	Sun et al. (2024)

RMSE: root mean square error, MAE: mean absolute error, NRMSE: normalized root mean square error, MAPE: mean absolute percentage error.

Table 2
The latest applications of ML in industrial-scale AD to predict biogas production.

Substrate	Input variables	Output variables	ML algorithm	Best accuracy	Reference
Grass silages, maize, pig manure, and cattle manure	15 variables including TS, VS, HRT, NH_4^+-N, OLR, VFA, detergent fiber, detergent fiber	Biogas production	ANN, GA, ACO	Prediction error: 6.24%, $R^2 = 0.90$	Beltramo et al. (2019)
Agricultural, municipal, and industrial biowastes	20 variables including input volume of kitchen food waste, chicken litter, percolate, and municipal fecal residue	Biogas production	SVM, XGBoost, RF, Logistic regression, KNN	KNN; $R^2 = 0.87$	de Clercq et al. (2019)
Agricultural, municipal, and industrial biowastes	18 variables including day number, input volume of various biowastes, co-digested wastes	Biomethane production	RF, Elastic Net, GBR, XGBoost	XGBoost; $R^2 = 0.88$	de Clercq et al. (2020)
18 types of biowastes, mainly seaweed, manure, eulat and pectin	Not provided	Methane production	Gompertz model, 15 commonly used ML model, Gompertz-ML	Gompertz-ML; MAPE = 4.52%	Hansen et al. (2020)
Sludge from the wastewater treatment plant	12 variables including 7 operation parameters and TP, COD, BOD_5, TN, TSS	Biogas yield	4 types of ANN: linear network, MLP, RBF and GRNN	$R^2 = 0.741$, deviation ratio: 0.697	Sakiewicz et al. (2020)
Sludge, thickened waste activated sludge, and 29 types of trucked wastes	Input volumes of 31 waste streams, TS, VS, VFA, ALK, VFA/ALK	Biogas yield	TPOT	$R^2 = 0.72$, RMSE = 247	Wang et al. (2021)
Sedimentation sludge, and food waste leachate	16 variables including sludge input/output amount, SRT, temperature, BOD, COD, VS/TS ratio, TN, TP, and SS	Biogas yield	LSTM, DA-LSTM, DA-LSTM-VSN	DA-LSTM-VSN; $R^2 = 0.76$, NRMSE = 0.09, MAE = 216.1	Jeong et al. (2021)
Food waste	19 variables including feed amount, feedstock TS, feedstock COD, TN, Oil, digester TS, digester COD, VFA, ALK, and VFA/ALK	Biogas yield	LR, RF, ANN, SVM, XGBoost	RF; $R^2 = 0.74$, RMSE = 0.51	Li et al. (2022a)
Food waste	14 variables including pH, TVFA, TS, VS, TAN, COD, pH, ALK, TS, VS, TAN, TVFA, COD, FV	Biogas yield	SMOTER-GA-ELM, ELM, GA-ELM, SMOTER-ELM	ELM; $R^2 = 0.972$	Zhang et al. (2022)
Manure, vegetable waste, and slaughterhouse waste	Operational parameters, TS, VS, temperature, pH, ALK and VFA	Biogas yield	RF, ANN, KNN, SVR, XGBoost	RF; $R^2 = 0.9242$, MAE = 1119.60, RMSE = 1405.84	Yildirim and Ozkaya (2023)

RMSE: root mean square error, MAE: mean absolute error, NRMSE: normalized root mean square error, MAPE: mean absolute percentage error.

variables, achieving an optimal calibration model with an R^2 of 0.76. Subsequently, Bai et al. (2017) also employed LR methods and software sensor technology for ALK prediction. They identified that, among acidity–alkalinity, ORP, and EC (used as input variables), EC significantly influenced ALK prediction. However, LR or MLR models can encounter overfitting or instability issues, especially with high-dimensional data or in situations of multicollinearity, leading to inaccurate predictions (Todeschini et al., 2004). To overcome these limitations and improve predictive performance, more complex and accurate ML algorithms have been applied to ALK prediction. For instance, Wang et al. (2018) used an ANN to track ALK changes during AD processes online. On the basis of existing online measurement parameters, they established a 3-2-1 structured backpropagation–feedforward neural network model with a high R^2 value of 0.9948. Sensitivity analysis further highlighted ORP as the most significant factor.

However, monitoring a single indicator typically struggles to comprehensively assess the operational state of AD digesters. To gain a more comprehensive understanding of digester operation, this typically necessitates the integration of complex systems involving online sensors and ML algorithms to simultaneously monitor the changing trends and relationships among multiple indicators. Jia et al. (2022) investigated deep learning models for predicting online sensor data related to state and performance variables. They found that a combined CNN and LSTM model outperformed individual CNN and LSTM models in predicting stability and performance variables (such as pH, alkalinity, VFA, COD, SCOD, TS, VS, methane production rate, and methane content in biogas) with high accuracy ($R^2 = 0.978$; mean square error = 0.031). Wang et al. (2024) integrated a millielectrode array physical sensor (MAPS), an AD simulation system (ADSS) based on AD Model No.1, and explainable ML algorithms for real-time *in situ* monitoring of pH, ORP, temperature, NH_4^+-N, and EC. Compared with conventional ML algorithms, the integrated MAPS–ADSS–ML method achieved 99% prediction accuracy. Furthermore, the adverse effects of pH, alkalinity, and NH_4^+-N on biogas production were confirmed by their inverse relationship with VFA, as indicated by SHAP values and the predicted negative correlation coefficients of Pearson ($r = -0.93$) and Spearman ($rs = -0.92$).

Significant developments have been made in the application of ML for the simultaneous monitoring of single or multiple indicators. However, the environment of AD systems tends to shorten the lifespan of physical sensors, thereby hindering the acquisition of long-term data. Moreover, existing ML algorithms typically rely heavily on statistical data while neglecting the underlying theories and principles of organic waste and biological systems, resulting in a disconnect between model predictions and physicochemical explanations. Therefore, the development of high-precision, long-life AD monitoring sensors, along with the exploration of integrated approaches combining physicochemical models and ML, has emerged as a critical avenue for future research to digitalize and comprehensively interpret AD systems.

3.3. Optimization of parameters

Optimized substrate ratios, suitable temperature and pH ranges, balanced C/N ratios, and appropriate concentrations of VFA and ALK are crucial factors in enhancing substrate digestion and methane production (Bong et al., 2018; Khan et al., 2016; Matheri et al., 2018). Conventional parameter optimization methods typically involve laboratory batch experiments to optimize various operational parameters to increase substrate digestion and methane yield (Kunatsa & Xia, 2022; Ramachandran et al., 2019). However, this approach has drawbacks such as complexity, time and labor intensiveness, and high costs. To overcome these challenges, researchers have adopted various ML algorithms to optimize substrate ratios or process parameters, thereby achieving comprehensive optimization of the AD process and maximizing biogas and methane production.

Gray relational analysis–PCA (GRA–PCA), central composite design–RSM (CCD–RSM), and ANNs are effective in optimizing AD processes. For example, Senthilkumar et al. (2016) used GRA–PCA to determine the best process parameters for AD of food waste, finding optimal conditions at 7.5% TS, pH 7, 50 °C, and 30% poultry waste co-digestion to achieve the highest biogas production. Jiménez et al. (2015) employed CCD–RSM to conduct anaerobic co-digestion experiments, optimizing conditions for increased methane yield. The optimized specific methane yields under mesophilic and thermophilic conditions were 1.31 and 1.38 g CH_4-COD $gVSS^{-1}$ d^{-1}, respectively. Kainthola et al. (2020) optimized co-digestion parameters for agricultural and food waste using CCD–RSM. Validation experiments under an optimized C/N ratio of 30, pH of 7.32, and F/M ratio of 1.87 exhibited methane production of 323.78 mL/g-VS_{add}, which is a 166.54 mL/g-VS_{add} increase over single digestion. These studies highlight that GRA–PCA and CCD–RSM can effectively reduce the number of experiments, aiding in determining the best waste combination and process parameters. Dahunsi et al. (2016) further compared the optimization effects of ANNs and RSM for the co-digestion of poultry manure and papaya peel. The ANN model achieved a higher R^2 value (0.9828) than the RSM model (0.9181), indicating that the ANN model was more accurate and efficient.

4. ML applications in composting

Composting technology represents a sustainable approach for biologically treating organic waste, primarily focusing on organic components that are easily biodegradable by aerobic bacteria, such as cellulose, proteins, and fats. The resulting compost not only functions as an organic fertilizer to improve soil quality and diminish dependence on chemical fertilizers but also acts as a soil amendment, reducing heavy metal bioavailability, tackling organic pollutants, and mitigating soil pathogens. However, conventional composting technology still has drawbacks, including low humification rates, greenhouse gas emissions, nitrogen loss, and high metal dispersion (Chen et al., 2023; Guo et al., 2020; Li et al., 2022c). Low humification during the composting process restricts the use of the resulting compost in agricultural applications (Bernal et al., 2009; Dias et al., 2010). Nitrogen loss, greenhouse gas emissions, and the dispersion of heavy metals throughout the composting process not only deteriorate the quality of the compost but also significantly contribute to environmental contamination (Chen et al., 2019; Yang et al., 2020). To overcome these limitations and address these deficiencies, an increasing number of researchers are exploring the use of ML in monitoring and controlling composting processes to enhance product quality, optimize the composting processes, and better protect the environment (Aydın Temel et al., 2023; Wan et al., 2022; Wang et al., 2023a). Currently, in the limited available literature, ML algorithms, such as ANN, RF, SVM, ANFIS, and KNN, have been used to predict compost maturity and other critical parameters and monitor pollution and greenhouse gas emissions, among other aspects. The application principles of ML-based studies for composting are presented in Fig. 3.

4.1. Prediction of composting maturity and optimization of parameters

Partially decomposed compost contains harmful bacteria, acidic substances, and organic or inorganic pollutants that impede plant growth, contaminate the surrounding environment, and ultimately pose threats to both ecosystems and human health (Wan et al., 2022; Xue et al., 2019). Therefore, it is essential to guarantee the

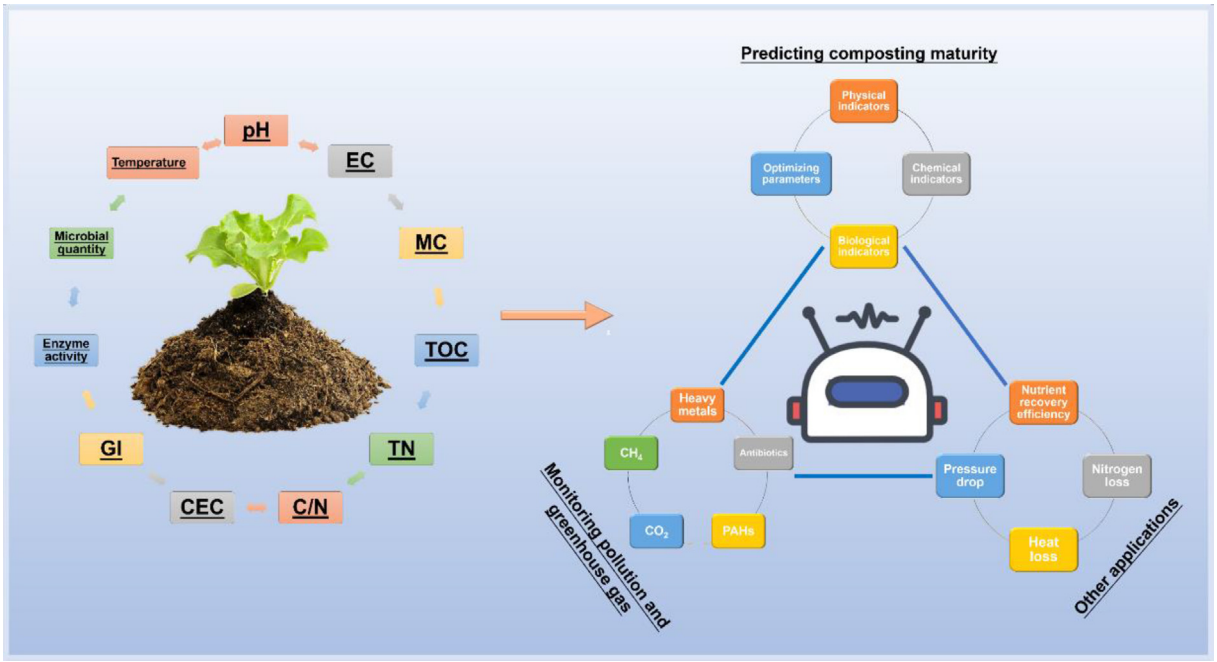


Fig. 3. Principles of ML-based studies for composting technology.

maturity of organic waste compost, as it significantly impacts both the composting process and the final compost product quality. Key parameters for assessing compost maturity include physical indicators (temperature, color, odor, etc.), chemical indicators (pH, EC, C/N ratio, moisture content (MC), humic acid content, total organic carbon (TOC), etc.), and biological indicators (seed germination index (GI), enzyme activity, microbial quantity, etc.) (Aydın Temel et al., 2023; Guo et al., 2021a). Table 3 summarizes the recent research on the application of ML models in predicting compost maturity.

4.1.1. Physical morphology and color changes to assess maturity

During the composting process, the decomposition and humification of organic matter change its color to a dark brown or black hue. Consequently, by observing the physical morphology and color changes, the maturity of the compost can be roughly assessed. To enhance both the efficiency and accuracy of compost maturity prediction, researchers have integrated the CNN model, which involves direct and in-depth analysis of compost images for assessing compost maturity. Xue et al. (2019) constructed a CNN model to predict compost maturity by processing images through convolutional pooling layers. This method accurately captured granularity differences in compost images across different stages and precisely extracted visual features, enabling the identification of compost maturity. Kujawa et al. (2020) explored the feasibility of early maturity classification of compost using CNNs. The study found that the CNN model effectively classified early compost maturity, with classification errors ranging from 0.51% to 17.77%. Sangeetha and Govindarajan (2023) employed a rapid region-based CNN prediction model to extract multilayer image attributes from different compost stages for rapid maturity assessment, with an accuracy rate of 96.4%. Overall, the introduction of CNN models for compost maturity identification significantly improved prediction efficiency and accuracy, surpassing the 95% accuracy threshold. This image-based identification method aligns closely with practical applications, undoubtedly providing robust technical support and a reference for accurately determining compost maturity in production practices.

4.1.2. Chemical indicators to assess maturity

Temperature, pH, MC, EC, total nitrogen (TN), C/N ratio, and TOC are significant chemical indicators for assessing compost maturity (Aydın Temel et al., 2023; Guo et al., 2021b). Bayındır et al. (2022) used a cascade forward neural network (CFNN) model, with the cattle manure ratio and composting time as input variables, to predict maturity parameters such as temperature, pH, EC, MC, TN, C/N ratio, and TOC. Their results showed that the manure ratio significantly affects the composting process. Using GAs for optimization, the optimal manure ratio and processing time were 23.39% and 104.86 days, respectively. Yılmaz et al. (2022) used the radial basis functional neural network (RBFNN) to investigate the impact of tea waste ratio and composting time on parameters such as MC, TN, and TOC loss in the cocomposting process of food waste and tea waste. The RBFNN model achieved the lowest TN and TOC loss in the composting process with a 25% tea waste ratio. Further optimization of co-composting process parameters, including

Table 3
The latest applications of ML to predict composting maturity.

Biomass	Input variables	Output variables	ML algorithm	Best accuracy	Reference
Vegetable waste, straw, livestock waste and mixtures	Image of composting	Composting maturity	CNN	99.7%	Xue et al. (2019)
Sewage sludge, rapeseed straw	Image of composting	Composting maturity	CNN	Classification error ranged from 0.51% to 17.77%	Kujawa et al. (2020)
Grass, livestock-manure, vegetable-wastes, plastic, glasses	Image of composting	Composting maturity	R−CNN	96.4%	Sangeetha and Govindarajan (2023)
Municipal solid waste and cow dung	Cattle manure ratio, composting time	pH, EC, MC, TN, C/N ratio, TOC	CFNN, RSM, GA	CFNN-GA; Total desirability was 0.4455	Bayındır et al. (2022)
Food waste and tea waste	Tea waste ratio, composting time	MC, TN, TOC loss	RBFNN, GA	Desirability levels are above 95 %	Yılmaz et al. (2022)
Olive mill waste and natural mineral amendments	Natural mineral material types, olive mill waste ratio, composting time	Temperature, pH, EC, MC, TN, and C/N ratio	FF-NN, ER-NN, RSM, GA	Prediction error <2%	Aycan Dümenci et al. (2021)
Traditional chinese medicine residues	C/N ratio, initial moisture content, inoculant type, composting days	Humic acid content	ANN, GA	$R^2 = 0.91$	Shi et al. (2022)
Chicken manure, corn straw	Penicillin, pH, and C/N ratio	Bacterial communities and humification processes	RF and Linear regression	–	Kang et al. (2022)
Textile waste, green waste, paper & cardboard waste	UV−visible spectroscopy, X-ray diffraction, and infrared spectroscopy	Humification degree, presence of cellulose mixtures, aromaticity, and polymerization degree	PLS	–	Biyada et al. (2020)
Rice straw, dried pig manure	NIRS figures	Organics, TN, C/N ratio	PLS	–	Shen et al. (2023)
638 datasets about green waste composting	Composting time, temperature, pH, EC, C/N ratio, TN, OM, E4/E6, humic acid, TOC, the presence of additives	GI, T-value	Extra Trees, GBR, MLP, KNN	Extra Trees; $R^2 = 0.928$ for GI, $R^2 = 0.957$ for T-value	Li et al. (2023)
Food waste, garden waste	Composting time, ambient temperature, ventilation rate, pH. EC	GI, C/N ratio	RF, XGBoost, LightGBM, MLP	XGBoost; $R^2 = 0.977$ for C/N ratio, $R^2 = 0.986$ for GI	Wan et al. (2022)
452 kitchen waste composting samples	Composting time, temperature, MC, pH, C/N, ammonia nitrogen, nitrate nitrogen, TN, TOC, GI, EC, organic matter	Composting maturity	Linear regression, KNN, DT, SVR, RF, Stacking model	Stacking model; Training $R^2 = 0.959$, testing $R^2 = 0.875$	Ding et al. (2022)

temperature, pH, EC, C/N, MC loss, TN loss, and TOC loss, has been performed using GAs. Aycan Dümenci et al. (2021) predicted compost maturity parameters (temperature, pH, EC, MC, TN, and C/N ratio) using a neural network model with natural mineral materials, olive waste ratio, and composting time as inputs. The model had prediction errors below 2%. After GA optimization, blending pumice and vermiculite achieved the desired temperature and reduced nitrogen loss with pH, EC, and MC adjustments. In addition, ML combined with sensors was used to monitor MC in industrial-scale composting systems (Moncks et al., 2022).

Humic acid content is also a crucial chemical indicator for evaluating compost maturity and quality. As the composting process proceeds, microorganisms break down organic matter to produce metabolites, including humic acid, which accumulates to indicate compost maturity when it reaches a stable level (Guo et al., 2021b). Shi et al. (2022) predicted humic acid content in compost with a backpropagation neural network model using composting parameters (C/N ratio, initial MC, inoculant type, and composting days), achieving an R^2 value of 0.91. Through GA optimization, they identified the best parameter combination (C/N ratio of 37.42, MC of 69.76%, sterile conditions, and 50 days of composting). Under these conditions, the compost products had the highest humic acid content. Kang et al. (2022) investigated the impact of penicillin, pH, and C/N ratio on bacterial communities and humification processes in aerobic composting using RF and LR models. Their results indicated that penicillin primarily influenced low-abundance microbial genera, with bacterial communities regulating the synthesis of humic substances, including fulvic acid and humic acid. Quorum sensing modulates carbohydrate and amino acid metabolism, controls humic acid synthesis, and influences bacterial cooperation during composting.

Spectroscopy and other chemical characterization methods are crucial tools for predicting compost maturity through ML. Vergnoux et al. (2009) initially used near-infrared spectroscopy (NIRS) to track physicochemical and biochemical changes during composting. Using PCA, they identified distinct compost stages and aging processes based on spectral data. The incorporation of PLS models facilitated the prediction of diverse parameters, including MC, temperature, pH, organic carbon (OC), organic nitrogen (ON), OC/ON ratio, TN, NH_4^+-N, and humic and fulvic acid ratios. Biyada et al. (2020) examined various chemical changes, such as humification degree, cellulose mixtures, aromaticity, and polymerization degree, during the composting process using ultraviolet–visible spectroscopy, X-ray diffraction, and infrared spectroscopy. Moreover, through the application of PLS models, they further analyzed the correlation between infrared spectra and compost chemical properties. Shen et al. (2023) performed quantitative analysis of organics, TN, and the C/N ratio under different composting conditions using NIRS. They constructed three nonlinear PLS models employing spectral preprocessing and variable selection strategies. The results indicated that with appropriate spectral preprocessing, NIRS demonstrated accurate predictions for each element, exhibiting notable predictive capabilities.

4.1.3. Biological indicators to assess maturity

The seed GI comprehensively reflects the phytotoxicity of compost and is considered the most sensitive and reliable indicator of compost maturity (Komilis & Tziouvaras, 2009; Yang et al., 2021b). Simultaneously, enzyme activity, microbial quantity, and other related indicators reflect the quantity and activity level of microorganisms in compost, enabling an assessment of its maturity (Goyal et al., 2005; Lv et al., 2022). Li et al. (2023) used extra trees, GBR, MLP, and KNN to predict the GI and T-value. The extra tree algorithm achieved the best prediction accuracy (R^2 of 0.928 for GI and 0.957 for the T-value). Pearson correlation analysis revealed that humic acid notably impacts the GI, whereas the C/N ratio during the composting process significantly affects the T-value. In another study, Wan et al. (2022) employed RF, XGBoost, LightGBM, and MLP to forecast both the GI and C/N ratio. XGBoost emerged as the most effective in predicting the C/N ratio, with an R^2 value of 0.977. However, regarding GI prediction, the ensemble model performed the best, achieving an R^2 value of 0.986. Feature importance analysis highlighted the considerable influence of different seed types on the GI, and the physicochemical properties of the raw materials significantly affected the prediction of the C/N ratio. Chakraborty et al. (2014) integrated visible and near-infrared (VisNIR) diffuse reflectance spectroscopy (DRS) with various ML models to forecast the enzymatic activity in compost. Among the evaluated models, such as PLS, PCA, RF, SVR, and PSL, the MLP–ANN model was the most accurate predictor, achieving an impressive R^2 value of 0.91, validating the effectiveness of combining VisNIR DRS with ML in forecasting compost enzyme and microbial activity.

4.1.4. Relationship between composting maturity and key parameters

Ding et al. (2022) used existing literature data to investigate the effects of crucial parameters (including composting time, temperature, MC, pH, C/N, NH_4^+-N, nitrate nitrogen, TN, TOC, GI, EC, and organic matter) on compost maturity, which was evaluated through a maturity score using ML models. Compared with LR, KNN, DT, SVR, and RF, the stacking model exhibited superior predictive accuracy (with a training R^2 value of 0.959 and a testing R^2 value of 0.875). Simultaneously, SHAP and partial dependence analysis revealed that the mesophilic phase should maintain appropriate temperature, MC, and pH values, whereas the thermophilic phase should be fine-tuned on the basis of partial dependence analysis to provide suitable nutrients. Optimal initiation conditions should maintain a mesophilic environment with a temperature range of 30°C–45 °C, an MC between 55% and 65%, and a pH level of 6.3–8.0. In addition, during the thermophilic stage, it is essential to ensure that the TN content exceeds 2.3% and the TOC content exceeds 35%. Experimental validation confirmed the improved maturity achieved by the optimized model-based systems.

4.2. Monitoring of pollution and greenhouse gas emissions

Currently, research on the prediction and control of pollutants in the composting process using ML remains limited. Specifically, ML applications in composting pollutant control mainly concentrate on heavy metal immobilization and organic pollutant degradation. Guo et al. (2022) used various models with GAs to predict and optimize heavy metal stabilization in composting, with GBR performing best (R^2: 0.76–0.94 for the bioavailability trend and 0.55–0.81 for the immobilization rate). The key factors affecting bioavailability included initial metal bioavailability, total phosphorus, and composting time. After GA optimization, high immobilization rates were achieved for Cu (79.53%), Zn (31.30%), Cd (14.91%), As (46.25%), and Cr (66.27%).

The application of ML in monitoring and controlling organic pollutants, particularly antibiotics and polycyclic aromatic hydrocarbons (PAHs), has been a focal point. Alavi et al. (2019) used an ANN model to simulate the breakdown of tetracycline antibiotics in the composting process, achieving a remarkable R^2 value of 0.99. Furthermore, thorough sensitivity analysis revealed that time was the most significant factor, accounting for 80.43% of the impact, followed by the type of antibiotic (7.95%), the ratio of the bulking agent (6.43%), and the initial concentration of the antibiotic (5.17%). Wu et al. (2013) used six ML models to forecast bioavailability changes of 16 PAHs in compost-contaminated soil. The

performance ranking of the ML models was as follows: radial basis function > M5 model tree > SVR > MLP > M5 rules > LR. Key variables such as incubation time, OC, soil moisture, and nutrients significantly affect PAH bioavailability. Dragoi et al. (2021) combined ANNs and differential evolution algorithms to model and forecast the removal of petroleum hydrocarbons and OC during the composting of oily sludge. Their findings showed that the ANN model closely matched the experimental results, with a mean absolute relative error of 5.96% for hydrocarbons and 12.7% for OC. Other studies have successfully combined ANNs with various biosensors to precisely determine the concentrations of phenols, including phenol and catechol, generated during the composting process (Tang et al., 2008; Zhang et al., 2009; Zhou et al., 2014).

In addition, the application of ML mainly involves monitoring greenhouse gas emissions during the composting process. Díaz et al. (2012) conducted preliminary research using the ANFIS model to investigate factors affecting CO_2 emissions during composting. Their findings identified aeration time as the primary factor affecting CO_2 release. Under conditions of 0.2 Lair $(min \cdot kg)^{-1}$ aeration, 55% MC, and 1–3 cm particle size, the composting process exhibited the highest CO_2 emission levels. Li et al. (2022d) employed AdaBoost, bagging, gradient boost, RF, KNN, and DT to predict CO_2 generated during composting. The RF model achieved the highest classification accuracy of 88% and the lowest root mean square error (RMSE). TOC was identified as the most critical factor, contributing to 59% of CO_2 generation. Najafi and Ardabili (2018) explored the influence of C/N, reactor temperature, and retention time on methane production during composting. Their results indicated that methane production peaked at 35 °C with a C/N ratio of 20 and at 55 °C with a C/N ratio of 30. In addition, modeling the methane production process using logistic regression, ANFIS, and ANN models revealed that the ANFIS network accurately predicted methane production under both high and moderate temperature conditions. The above studies showcase the tremendous potential of ML in predicting greenhouse gas emissions from organic waste composting. However, there is limited research on the use of ML methods for monitoring and controlling other harmful gases (such as NH_3 and N_2O) during the composting process. Subsequent research can delve deeper into investigating and addressing this gap.

4.3. Other applications

ML not only plays a significant role in predicting compost maturity, optimizing process parameters, controlling pollution, and monitoring greenhouse gas emissions but also has shown vast potential in predicting nutrient recovery efficiency, nitrogen loss, heat loss, and pressure drop during the composting process.

Hosseinzadeh et al. (2020) evaluated the performance of ANN and MLR models in predicting nutrient recovery efficiency in vermicomposting. Seven chemical and biological indicators were used as input variables to predict the recovery rates of TN and total phosphorus. The results demonstrated a significant increase in the recovery rates of TN and total phosphorus through composting. The ANN model outperformed the MLR model in predicting TN and total phosphorus, with TN and C/N ratio being key factors in the ANN predictions.

Kabak et al. (2022) investigated the impact of physicochemical parameters on nitrogen loss during the composting process. The findings revealed that nitrogen loss increased with increasing temperature and was influenced by factors such as pH, compost mixing, C/N ratio, and MC. Subsequently, they developed a linear and nonlinear hybrid tool based on CFNN to predict nitrogen loss, with the model demonstrating a mean absolute percentage error of approximately 1%–2%. Through GA optimization, the parameter values that minimize nitrogen loss were determined.

Boniecki et al. (2013) employed an ANN model to estimate heat loss during the composting process. The results indicate the effectiveness of the model in predicting heat release and loss throughout the composting cycle. Sensitivity analysis revealed the following six most crucial parameters in descending order: temperature, mineral mass, oxygen content, flow rate, CO_2 content, and process duration. These findings facilitate a better understanding and optimization of heat utilization in the composting process.

Sidełko et al. (2019) used an ANN model to predict the pressure drop of organic material beds during composting operations. The constructed MLP 5-9-1 neural network model exhibited a high correlation between the predicted and actual values. Variables such as time, density indicators, and hydraulic load, excluding flow direction, significantly influenced the prediction of airflow resistance. Statistical evaluation metrics revealed the high quality of the neural network, with an R^2 value of 0.906 and a standardized residual range of 4.082–5.453.

5. ML applications for insect treatment of organic waste

In recent years, insect farming technology has seen extensive use in organic waste treatment because of its economic viability, low operational costs, high social acceptance, and significant societal contributions (Ojha et al., 2020; Singh & Kumari, 2019). Insect farming commonly involves insects such as BSFs, maggots, cockroaches, mealworms, and compost worms, which convert organic waste into high-value insect protein feed and organic fertilizer products (Addy et al., 2021). Among these, BSF is the most commonly farmed insect, designated by the Food and Agriculture Organization of the United Nations as an edible resource insect. BSF larvae are rich in active proteins, insect fats, probiotics, antimicrobial peptides, and chitin and serve as a staple or supplementary feed for animals (Siva Raman et al., 2022; Surendra et al., 2020). To date, there have been limited reports on the application of ML in insect farming. The sparse reports available mainly focus on forecasting the expected weight of BSFs, optimizing the farming conditions of BSFs, controlling its lifecycle, and predicting the protein content of BSF using ML. The application principle of insect farming technology based on ML is illustrated in Fig. 4.

Arshad et al. (2023) explored the impact of time, larval quantity, aeration frequency, waste composition, and container surface area on the efficacy of BSF treatment of kitchen waste. The conditions for the BSF treatment of kitchen waste were optimized using an ANN model and life cycle assessment, achieving an accuracy rate of 99.5%. The optimal conditions included four daily aeration runs, 600 larvae per kilogram of waste, and a stratified feeding process, resulting in substantial larval weight gain (2.2%–6.5%) and significant reductions in kitchen waste (weight reduction of 70%–93% and volume reduction of 65%–85% within 15 days).

Muinde et al. (2023) used LR, ridge regression, DT, RF, bagging, AdaBoost, and XGBoost to determine the optimal BSF farming conditions. The RF model achieved an RMSE of 2.91 and an R^2 value of 80.9%, effectively predicting the final weight of BSF. The key factors included bed length, feed formulation, average larval count per bed, feed depth, and cycle time. This tailored prediction tool not only estimates the BSF yield but also guides production optimization.

Proietti et al. (2022) explored the real-time monitoring and prediction of various stages of the BSF lifecycle using RGB, infrared, and hyperspectral imaging technologies, along with computer vision algorithms for image and data collection (Proietti et al., 2022). They focused on estimating the size/weight and movement/vitality during the pupal stage, the movement/vitality and temperature anomalies during the larval stage, and the correlation between larval growth and the selected diet. Their study

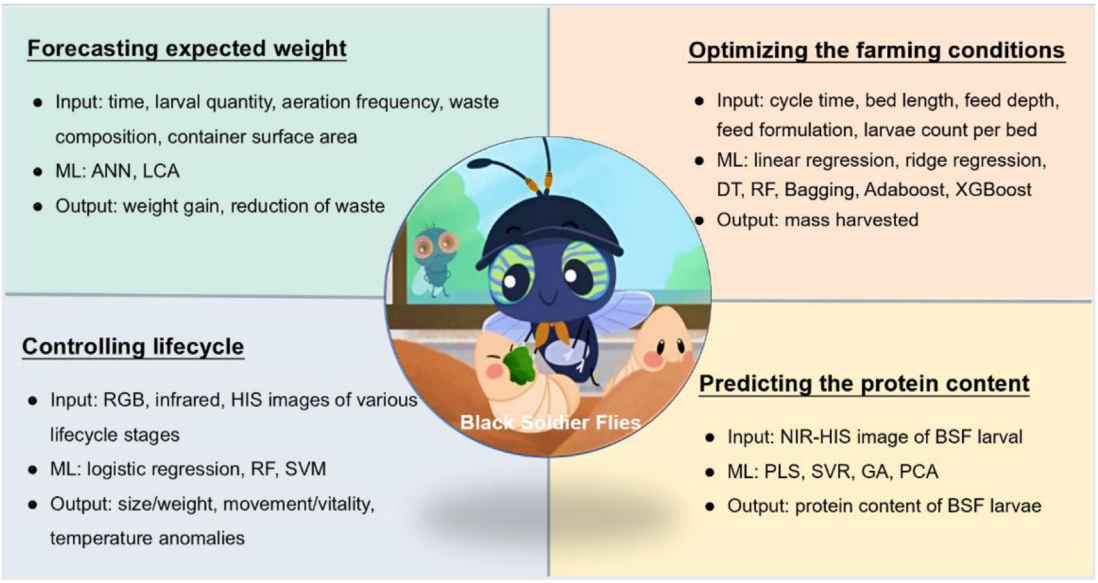


Fig. 4. Principles of ML-based studies on insect farming technology.

contributes to the enhancement of the control of the BSF lifecycle through noncontact measurements.

Cruz-Tirado et al. employed near-infrared hyperspectral imaging (NIR-HSI) in conjunction with PLS and SVR to develop a predictive model for the protein content of BSF larvae (Cruz-Tirado et al., 2023). The results indicated that both the PLS and SVR models exhibited strong predictive capabilities (RMSEP 1.57%–1.66%, RPD 2.0–2.5, and protein percentages 25.5%–43.5%). Furthermore, PCA illuminated the heterogeneous distribution of chemical compounds within larvae.

6. Challenges and potential prospects of machine learning applied in biological treatment of organic wastes

6.1. Potential challenges

Although ML demonstrates significant advantages and enormous potential in the biological treatment of organic waste, it also faces some potential challenges that may hinder its further development.

(1) Model Selection and Parameter Adjustment: Complex, variable, and unstable data in organic waste processing complicate the selection of an appropriate ML model. Different models exhibit varying sensitivities and adaptabilities, requiring a rigorous assessment of their strengths and weaknesses, along with careful experimentation and validation. Moreover, optimizing model performance requires the identification of ideal parameter combinations, which is challenging because of the extensive and intricate parameter space within the biological treatment of organic waste.

(2) Explainability and Reliability: ML models are typically perceived as “black boxes” because of their opaque decision making processes. This poses a significant challenge in the field of organic waste biological treatment, where comprehending the decision making process of a model is pivotal for ensuring its reliability. However, numerous ML models lack sufficient explainability, making it difficult for individuals to understand their predictive outcomes and decision logic. Therefore, enhancing the explainability and reliability of ML models is a critical research direction.

(3) Computational Resources and Time Costs: The training and inference processes of ML require extensive computational resources and time. This issue becomes particularly pronounced in the biological treatment of organic waste because of the vast amounts of data and the complexity of the models. Therefore, achieving efficient ML applications within limited computational resources and time constraints is a significant challenge.

(4) Lack of Practical Engineering Validation: Currently, the application of ML in biological treatment is mainly confined to the laboratory or pilot scale. Only through practical engineering validation can the applicability and performance of ML in biological treatment be more accurately evaluated, thereby promoting its widespread application and optimized development. However, practical engineering validation not only requires adequate resources and financial support but also necessitates close collaboration with the industry to jointly address potential issues and challenges.

6.2. Future perspective

The following approaches are considered potential avenues for further promoting the application of ML in the biological treatment of organic waste:

(1) Automated Model Selection and Parameter Tuning: Methods for automating model selection and parameter tuning can save considerable time and human resources. Furthermore, these methods can reveal high quality models and configurations that manual tuning may overlook by systematically exploring the model and parameter space. For instance, employing automated model selection algorithms, such as regularization paths, GA, or Bayesian optimization, to choose the most suitable model for the data; developing AutoML tools and frameworks such as TPOT or H_2OAutoML to automatically select appropriate models and optimize parameters based on data features and prediction objectives (Cheng et al., 2024).

(2) Research and Application of Explainable ML: To address the “black-box” nature of ML models, several explainable

frameworks have been developed to elucidate the contributions of input variables to the model, such as SHAP, LIME, Shapash, and InterpretML. However, only a few studies have adopted SHAP in AD (Gupta et al., 2023; Zhang et al., 2023). Therefore, these explainable methods should find broader applications in future research on organic waste biological treatment. In addition, human−computer interaction and feedback techniques appear to be viable paths for enhancing the explainability of ML models.

(3) Enhancing Computational Efficiency of Computing Platforms: The high computational resources and time required to process large amounts of data and complex models can be addressed using various strategies. For instance, leveraging the elastic scalability of cloud computing resources allows dynamic adjustment of computational power based on demand. The adoption of distributed computing enables the parallel processing of decomposed subtasks, thereby reducing the overall computation time. Furthermore, data compression and dimensionality reduction, incremental and online learning, and open-source tools and frameworks constitute effective avenues for achieving efficient ML applications.

(4) Strengthen Practical Engineering Application and Validation: To promote practical engineering applications and validation, specific measures can include i) conducting on-site experiments or long-term monitoring in biological treatment facilities to collect real-world data, applying ML models, and assessing their applicability and robustness in complex operational environments; ii) enabling data sharing and collaborative research using real-world biological processing facility data to accelerate the application and advancement of ML techniques, and leveraging lessons learned from successful implementations in other domains to refine methods for biological treatment; iii) developing customized ML solutions tailored to specific biological treatment facilities and treatment requirements, enabling better adaptation to actual operational environments and needs.

7. Conclusions

This review examines the applicability, advantages, and limitations of various ML models in the biological treatment of organic wastes. According to our findings, ANNs, tree-based models, SVM, and GAs have emerged as prominent algorithms extensively used in modeling biological treatment. Detailed analysis of ML applications in modeling AD, composting, and insect farming processes highlights the significant potential of ML in predicting products, optimizing processes, enabling real-time monitoring, and mitigating pollution emissions. However, despite significant strides, persistent challenges include complex parameter settings and model selection, the explainability and adaptability of algorithms, computational resource demands, and the lack of practical engineering validation. Future research should prioritize enhancing the efficacy of ML in biological treatment by automating model selection, employing explainable ML practices, enhancing the computing efficiency of the platform, and reinforcing practical engineering applications.

Author contributions

Long Chen: Conceptualization, Data curation, Formal analysis. **Pinjing He:** Investigation, Project administration, Resources, Supervision. **Hua Zhang:** Resources, Software, Supervision, Validation. **Wei Peng:** Supervision, Validation, Visualization. **Junjie Qiu:** Software, Validation, Visualization. **Fan Lü:** Funding acquisition, Project administration, Resources, Software, Validation, Visualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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